

Team Number: 44

Project Title: Breast Cancer Stage Classification with miRNA

Report Date: 9/15/25

Part one: Progress status as a team

All team members have detailed tasks listed on Teams Planner: YES or NO; if NO, explain.

Yes

Please answer the following questions:

1. Have you met as a team this past week? If yes, give date/time, and the members attended the meeting. If no, explain.

We didn't hold a formal team meeting, but we coordinated every day on our Discord server to share updates, assign tasks, and quickly resolve any questions or issues.

2. Have you met with the sponsor as a team? If yes, give date/time, and the member attended the meeting. If no, explain.

No, we had scheduled a team meeting with Dr. Ali Ibrahim for September 18th at 6 PM, but he had to postpone it due to faculty commitments.

3. Describe verbally the tasks completed the past week and the challenges faced.

During this week (Week 5) our team extended feature selection into deep learning to gain a broader view of which miRNAs are most informative for breast-cancer stage classification. Anika built and trained a simple Feedforward Neural Network (FNN) and also reviewed the miRNAs that appeared in both the FNN and her previous Sequential Feature Selection (SFS) results to confirm their support in current breast cancer research. Brehon implemented a Denoising Autoencoder (DAE) to extract a set of top-ranked miRNAs. Nicholas built a 1-D Convolutional Neural Network (1D-CNN) to rank key miRNAs and compared its selected features with those from his earlier chi-square analysis to show how the deep-learning approach complements the traditional method. Joshua developed an LSTM based sequential model to explore whether a recurrent architecture highlights different important features. Together, these models produced ranked miRNA subsets that complement the classical feature-selection results from last week and give a more complete picture of the most stable and biologically relevant features for breast-cancer stage classification.

4. Describe the tasks to be completed the coming week.

Next week (Week 6) our team will continue refining the deep learning feature selection work with additional input from Dr. Ali Ibrahim. Guided by his feedback, we plan to adjust or expand the neural network architectures we have implemented this week or explore variations such as deeper MLPs or alternative recurrent layers to improve performance and interpretability. After that, we are planning to compare the feature selection results across all deep learning approaches and the classical methods (SFS, chi-square, Variance Threshold, and LASSO) to identify a core set of miRNAs that consistently stand out.

Previous Report Summary. Date: 9/08/25

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
Joshua Adegboyo	2	0	2
Anika Baro Villa	2	0	2
Nicholas Campos	2	0	2
Brehon Parker	2	0	2

Current Report Summary. Date: 9/15/25

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
Joshua Adegboyo	2	0	2
Anika Baro Villa	2	0	2
Nicholas Campos	2	0	2
Brehon Parker	2	1	2

Name of the Member: Joshua Adegboyo

Report Date: 9/15/25

Task 4.1: Feature selection for relevant miRNAs identification (9/14/25)

During our meeting, Dr. Ali showed us various feature selection methods, which we would split amongst ourselves. These selections are Chi-square, NCA, and L1-norm. We will use these selections to reduce dimensionality and identify the most informative miRNAs for breast cancer stage classification. My focus was on the lasso feature selection method. Lasso feature selection was applied to the dataset to identify predictive features. However, with the chosen regularization, no features were selected.

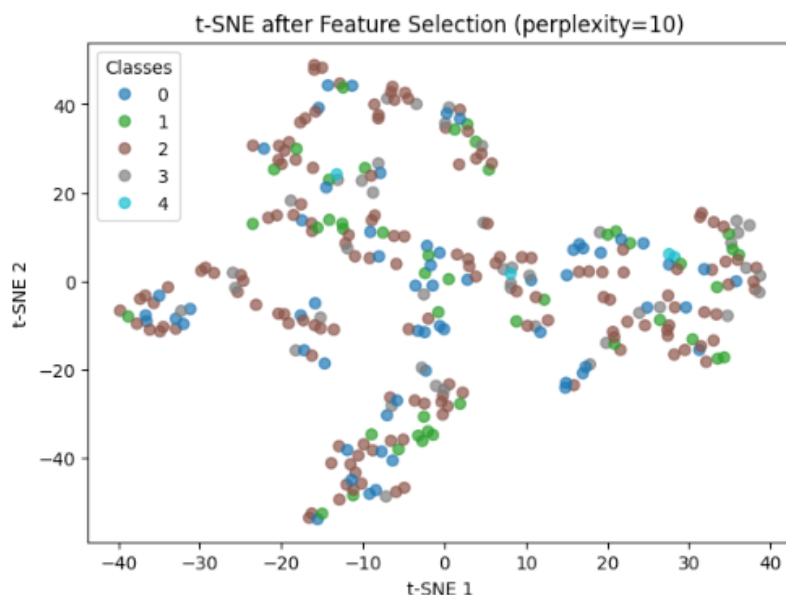
Task 4.2: Press Release Project (Logo, Contact information, Dateline and the Headline) (9/15/25)

For this project, we were tasked to put together a product press release that would include all standard sections, such as the logo, contact, dateline, headline, sub-headline, introduction, body, boilerplate, close, and team photos. We divided the sections across each team member, and my focus was on the Logo, Contact information, Dateline and the Headline.

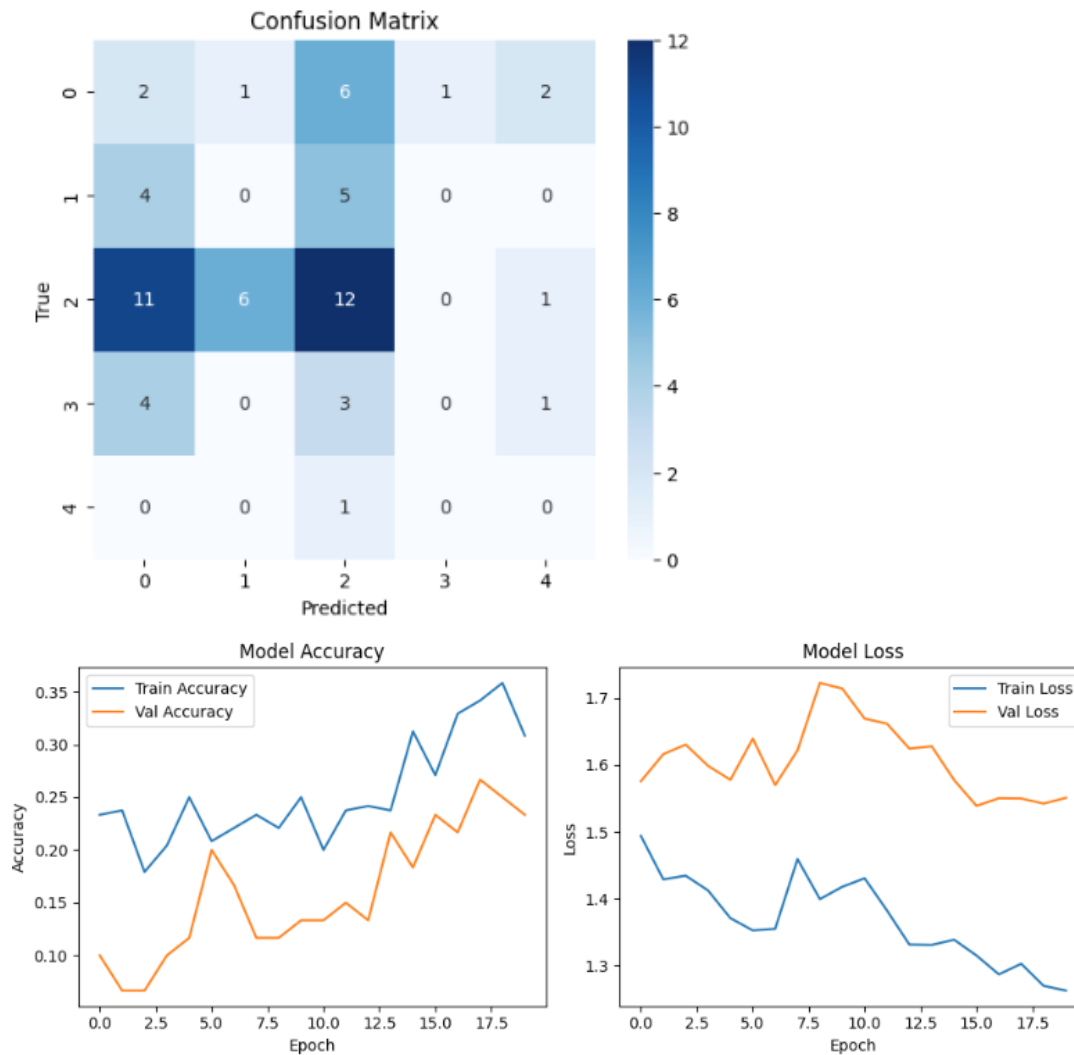
Task 5.1: Apply deep learning for feature selection (9/20/25)

I worked with the same breast cancer miRNA dataset (1,209 samples, 1,881 features, 5 classes). After standardization, I applied LassoCV and χ^2 feature selection, reducing the feature space to the top 50 miRNAs. I visualized the reduced space using t-SNE, which showed partial but imperfect stage separation.

For classification, I trained an LSTM-based sequential model in Keras, treating the 50 miRNAs as ordered features. The model included stacked LSTM layers with dropout and a softmax output. While LSTM did not outperform simpler models, this experiment highlights both the potential and the limitations of applying sequential deep learning to non-time-series omics data.



2/2 — 1s 589ms/step



Task 5.2: Review of Lasso-Selected and LSTM-Selected miRNAs

For this task I applied **LassoCV feature selection** in combination with **Chi² filtering** to reduce the high-dimensional miRNA dataset to a smaller subset of informative features. These reduced feature sets (e.g., top 50 miRNAs) were then modeled using an **LSTM-based sequential neural network**, which is well suited to capture dependencies among expression profiles. I also generated **t-SNE visualizations** to assess how well the selected features separate the different cancer stages.

From the Lasso/LSTM runs, I identified several candidate miRNAs that consistently appeared across reduced feature sets and contributed to classification, despite the challenge of class imbalance. I reviewed these candidates against the existing cancer biology literature and found that multiple miRNAs highlighted by the model already have known associations with tumor progression, signaling pathways, and therapy response, while others remain less studied. This provides both a validation of our modeling pipeline and a set of exploratory biomarkers for further analysis.

Task 6.1: Alternative deep learning models (9/28/25)

In addition to the baseline LSTM I trained, there is scope to test other recurrent or hybrid architectures (e.g., BiLSTM, GRU, or 1D-CNN–LSTM hybrids). These may capture more temporal dependencies in expression patterns and potentially highlight a different subset of stable miRNAs. If time allows, I'll also try a smaller architecture for efficiency, given the dataset's size.

Task 6.2: Cross-comparison and integration (9/28/25)

For this task, our next step is to pool the miRNA lists generated by FNN, SFS, CNN, DAE, and my LSTM results. The goal is to quantify overlap (e.g., Venn diagrams or intersection counts) and identify a consensus panel of miRNAs that consistently appear across methods. This combined feature set would be the strongest candidate for downstream validation and could serve as input for the final classifier model we present to Dr. Ali.

Name of the Member: Anika Baro Villa

Report Date: 9/15/25

Task 4.1: Apply Sequential Feature Selection (SFS) to identify the most informative miRNAs (9/12/25)

I applied Sequential Feature Selection (SFS) on our high-dimensional miRNA dataset. After reviewing the scikit-learn documentation and checking the class distribution, I chose macro-F1 scoring and `class_weight="balanced"` to handle the class imbalance. I built a scikit-learn Pipeline (StandardScaler followed by multinomial LogisticRegression) so that scaling occurred inside cross-validation and avoided data leakage. Because SFS is computationally heavy, I limited the runs (as Dr. Ali Ibrahim advised) to subset sizes $k = 10, 8, 5, 3$ with $cv=3$ and enabled parallelism. For each k I saved the selected miRNA list to CSV, plotted a t-SNE using only the selected features to check visual separability, and then produced an overlap summary. This analysis revealed a stable core of three miRNAs appearing in every list and five more appearing in at least two lists, giving the team a concise set of consistently significant features.

Task 4.2: Draft a product press release (sub-headline and introduction) (9/14/25)

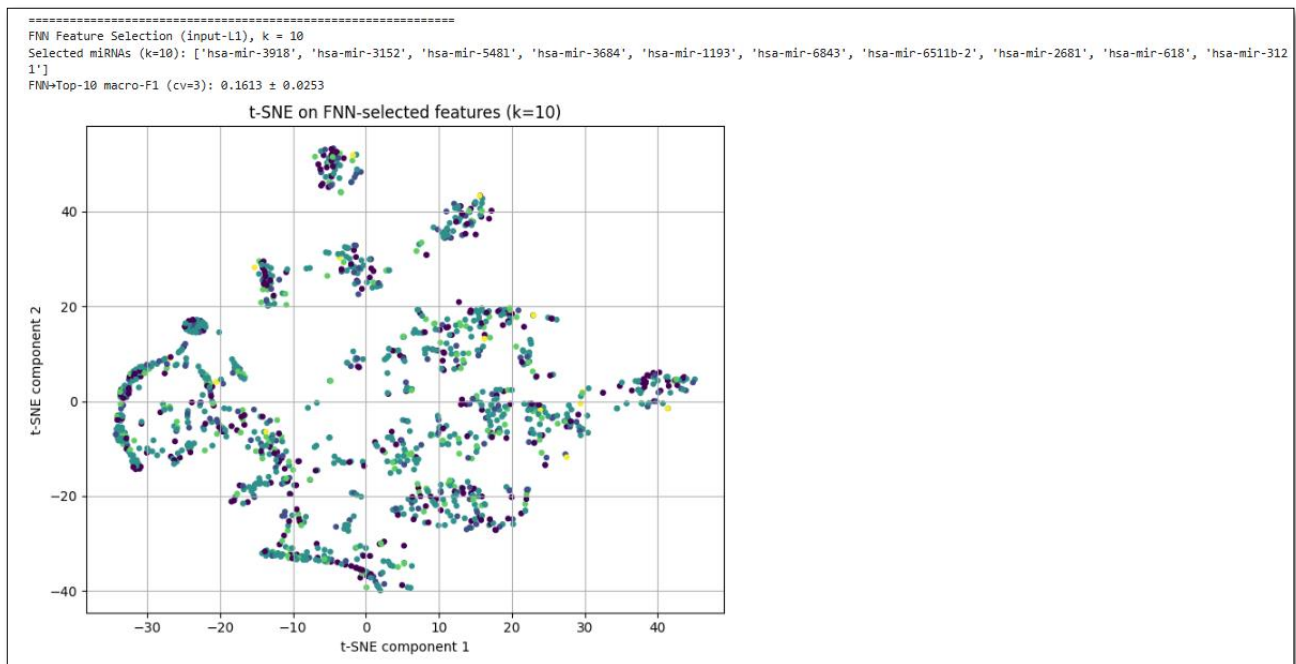
For this task, I worked on writing the sub-headline and introduction of our press release. The sub-headline was meant to get across the core benefit quickly, while the introduction pulled together the key details in a clear and engaging way. Additionally, I reviewed the sections my teammates drafted, offering suggestions to improve the document and made overall formatting changes to keep the press release consistent and compliant with the requirements.

Task 5.1: Apply deep learning feature selection with Feedforward Neural Network (FNN) (9/20/25)

For this task I reused the same setup as in the Sequential Feature Selection (SFS) analysis so the deep-learning results would be directly comparable. I used the same preprocessed miRNA dataset and class-balance check, applying macro-F1 scoring with `class_weight='balanced'` to give the minority stages proper weight. Then, I built a simple feed-forward neural network (FNN) in Keras with dense layers, dropout, ReLU activations and a softmax output for the five cancer stages (normal tissue plus stages I–IV). The input features were log-transformed and standardized, the random seed fixed for reproducibility, and the model trained with a validation split and early stopping. After training, I ranked miRNAs by the mean absolute value of their input-layer weights and saved the full ranking to `fnn_ranked_features.csv`.

From this ranking I extracted top- k feature sets ($k = 10, 8, 5, 3$). For each k I saved the selected miRNAs to CSV, evaluated their predictive power with a downstream multinomial Logistic Regression using 3-fold cross-validation and macro-F1 scoring, and produced t-SNE plots to visualize the separation of cancer stages. Additionally, I generated an overlap summary showing three miRNAs present in every top- k list and five more appearing in at least two lists, providing a concise consensus of consistently important features. The screenshots below show the saved CSV files, an example t-SNE plot for $k = 10$, and the overlap summary.

 fnn_k3_features	9/20/2025 3:48 PM	Microsoft Excel C...	1 KB
 fnn_k5_features	9/20/2025 3:48 PM	Microsoft Excel C...	1 KB
 fnn_k8_features	9/20/2025 3:48 PM	Microsoft Excel C...	1 KB
 fnn_k10_features	9/20/2025 3:48 PM	Microsoft Excel C...	1 KB
 fnn_ranked_features	9/20/2025 3:48 PM	Microsoft Excel C...	50 KB



```
=====
FNN overlap summary (k = 10, 8, 5, 3):
Common to ALL (10 n 8 n 5 n 3): ['hsa-mir-3152', 'hsa-mir-3918', 'hsa-mir-5481']
Appear in ≥2 of {10, 8, 5, 3}: ['hsa-mir-1193', 'hsa-mir-2681', 'hsa-mir-3152', 'hsa-mir-3684', 'hsa-mir-3918', 'hsa-mir-5481', 'hsa-mir-6511b-2', 'hsa-mir-6843']
```

Task 5.2: Perform literature review of stable miRNAs selected by SFS and FNN (9/20/25)

I took a closer look at the 16 miRNAs that showed up in at least two k-runs for both the Sequential Feature Selection and the Feedforward Neural Network methods. I wanted to see which of these repeatedly selected miRNAs already have solid scientific backing and could be dependable features for our classification model. Drawing on peer-reviewed journal articles such as Nature Cell Biology (Martello 2010 on miR-103/107) and Journal of Translational Medicine (Ji 2023 on miR-1183), I examined what is known about each miRNA's cellular functions, its links to cancer (especially breast cancer) and any reported or potential clinical applications. Out of the 16, nine are well documented with strong experimental evidence and clear connections to major cancer pathways such as PI3K/AKT/ERK signaling, cell-cycle control, apoptosis, EMT/metastasis, and therapy response. Three more have some early but limited data, mostly from high-throughput or initial prognostic and drug-resistance studies. The last four are still pretty much unexplored and so far only show up in sequencing datasets or predictive databases.

Task 6.1: Experiment with additional deep learning architectures (9/28/25)

I plan to extend the baseline FNN by testing at least one alternative deep-learning model using the same cleaned and labeled miRNA dataset. The goal is to see whether different network types change classification performance or highlight a different set of informative miRNAs. I'll record the training setup and key evaluation metrics such as macro-F1 and save the list of miRNAs each model identifies as most predictive.

Task 6.2: Perform cross-compare and integrate feature lists (9/28/25)

As a team we plan to compare the miRNAs highlighted by each deep-learning model, including the baseline FNN and other architectures such as a deeper MLP, CNN, or RNN/LSTM, with the feature lists from the classical methods (SFS, chi-square, Variance Threshold, LASSO). Our goal would be to identify a core set of miRNAs that consistently stand out across all approaches.

Name of the Member: **Nicholas Campos**

Report Date: **9/15/25**

Task 4.1: Chi2 Feature Selection Implementation and Analysis (9/11/25) I successfully implemented Chi-squared feature selection using a systematic loop-based approach to evaluate 10, 8, and 3 features for breast cancer stage classification. Using scikit-learn's SelectKBest with chi2 scoring function, I applied a for loop to iterate through each feature count (for k in [10, 8, 3]), ensuring consistent preprocessing with data normalization to handle the non-negative requirement for Chi2 testing. For each iteration, I applied the feature selection, transformed the data, and generated t-SNE visualizations to assess clustering patterns across cancer stages. The analysis revealed that 10 features provided the best visual separation between cancer stages compared to 8 and 3 features.

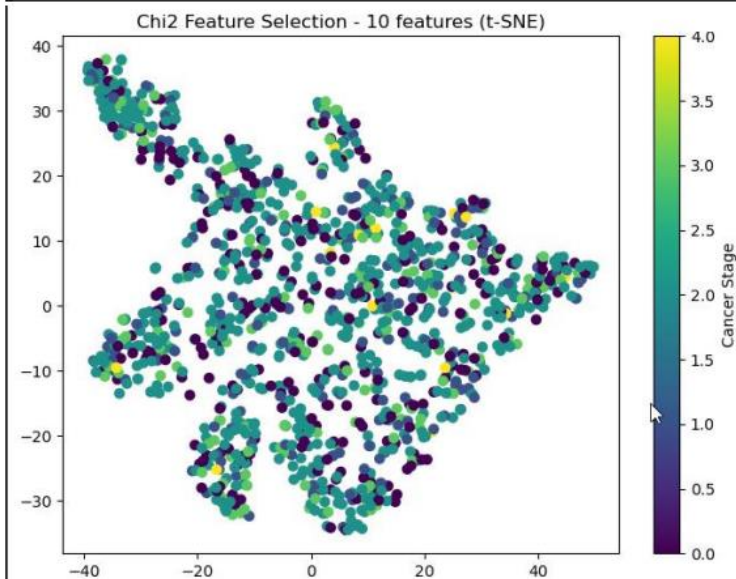
```
# Feature selection (Chi2, try 10 features, 8 features, 3 features and plot it using tsne)
from sklearn.feature_selection import SelectKBest, chi2

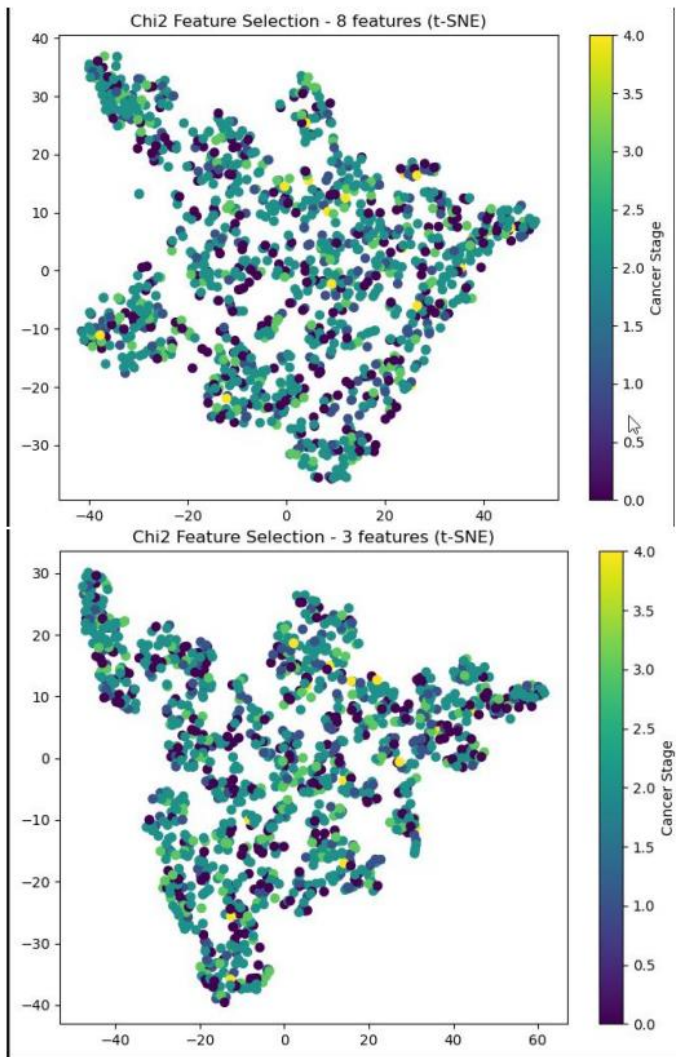
# Make data non-negative for chi2
data_positive = data - data.min() + 1e-10

# Chi2 feature selection with 10, 8, 3 features
for k in [10, 8, 3]:
    # Apply Chi2 feature selection
    chi2_selector = SelectKBest(score_func=chi2, k=k)
    data_selected = chi2_selector.fit_transform(data_positive, label)

    # Apply t-SNE
    tsne = TSNE(n_components=2)
    tsne_result = tsne.fit_transform(data_selected)

    # Plot t-SNE
    plt.figure(figsize=(8, 6))
    plt.scatter(tsne_result[:, 0], tsne_result[:, 1], c=label, cmap='viridis')
    plt.title(f'Chi2 Feature Selection - {k} features (t-SNE)')
    plt.colorbar(label='Cancer Stage')
    plt.show()
```





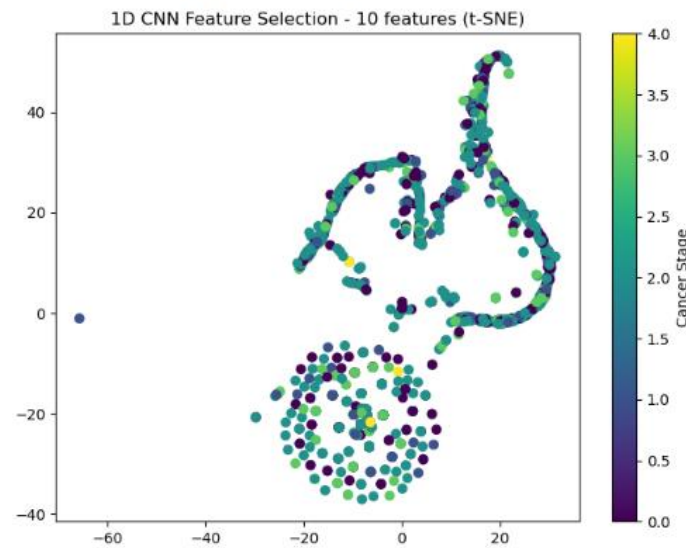
Task 4.2: Product Press Release Finalization and Team Photos Integration (9/11/25) I developed the comprehensive "Key Features" bullet points section highlighting our system's technical capabilities including advanced miRNA expression analysis, comparative ML/DL evaluation, rigorous data splitting protocols, and multiple feature selection methods. I also created the "Meet the Team" page with professional photos and names for all team members using proper formatting and layout design.

Task 5.1: 1D CNN Feature Selection Implementation (9/18/25)

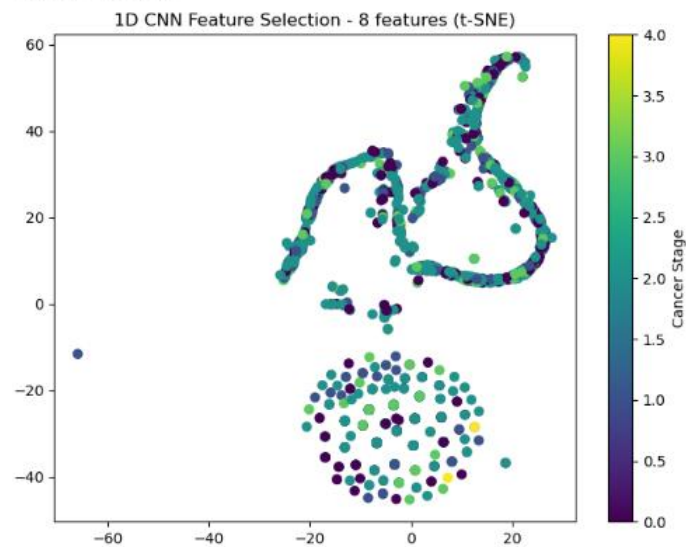
I successfully implemented a 1D Convolutional Neural Network approach for deep learning-based feature selection to complement my traditional Chi2 statistical method from the previous week. The CNN architecture consisted of three convolutional layers with 64, 32, and 16 filters respectively, followed by global max pooling and dense classification layers for cancer stage classification. Feature importance was determined by analyzing convolutional layer weights combined with data variance patterns, providing a learned representation approach rather than statistical independence testing. Using the same dataset and 70/15/15 split from my Chi2 analysis, I applied the CNN feature selection for 10, 8, and 3 features using a systematic loop-based approach similar to my previous implementation. The CNN training curves demonstrated successful convergence with validation accuracy reaching approximately 0.85 across different feature counts, and loss curves showed proper learning without significant overfitting, validating the model's ability to learn meaningful representations from the miRNA data. The approach generated t-SNE visualizations for each feature count and produced a ranked list of top miRNA features based on CNN importance scores. The

deliverable includes trained CNN models with documented feature importance rankings and selected miRNA biomarkers ready for comparative analysis against traditional statistical methods.

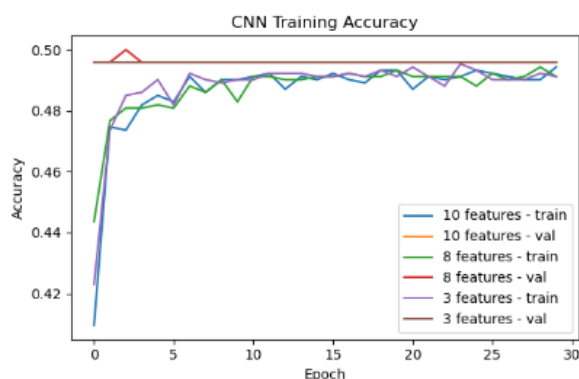
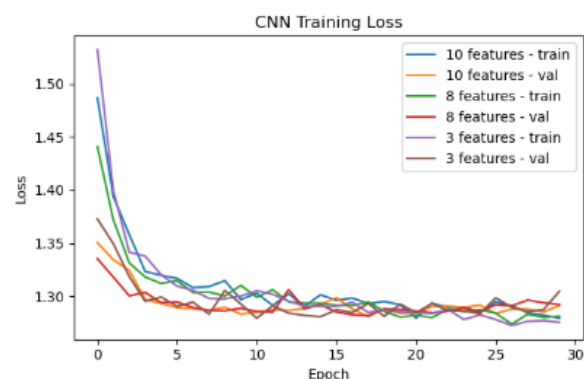
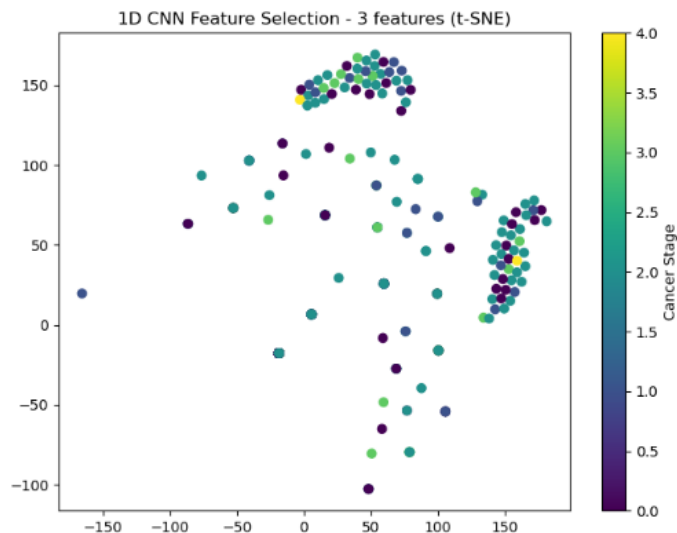
Selected feature indices: [1526 1429 1440 1184 80 1199 1188 764 1844 43]
CNN importance scores: [0.18889073 0.18889073 0.18889073 0.18889073 0.18889073 0.18889073 0.18889073 0.18889073 0.18889073 0.18889073]



Selected feature indices: [1440 1184 80 1199 1188 764 1844 43]
CNN importance scores: [0.18546296 0.18546296 0.18546296 0.18546296 0.18546296 0.18546296 0.18546296 0.18546296 0.18546296 0.18546296]



Selected feature indices: [764 1844 43]
 CNN importance scores: [0.19413034 0.19413034 0.19413034]



0	mirna_43	0.198890	0	hsa-mir-43	0.182997
1	mirna_1844	0.198890	1	hsa-mir-1844	0.182997
2	mirna_764	0.198890	2	hsa-mir-764	0.182997
3	mirna_1188	0.198890	3	hsa-mir-1188	0.182997
4	mirna_1199	0.198890	4	hsa-mir-1199	0.182997
5	mirna_80	0.198890	5	hsa-mir-80	0.182997
6	mirna_1184	0.198890	6	hsa-mir-1184	0.182997
7	mirna_1440	0.198890	7	hsa-mir-1440	0.182997
8	mirna_1429	0.198890	8	hsa-mir-1429	0.182997
9	mirna_1526	0.198890	9	hsa-mir-1526	0.182997

Top 10 preview:

	mirna	cnn_importance
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Task 5.2: Traditional vs Deep Learning Feature Selection Comparative Analysis (Chi2 vs 1D CNN) (9/18/25)

Building on both my Chi2 implementation and 1D CNN feature selection from the previous week, I conducted a systematic comparison framework evaluating traditional statistical versus deep learning approaches across multiple criteria. The analysis revealed notable differences in clustering performance, with Chi2 feature selection consistently producing more compact and well-separated clusters in t-SNE visualizations across all feature counts, particularly evident when comparing the 10-feature and 3-feature plots where distinct cancer stage groupings were clearly visible. The 1D CNN results showed more dispersed clustering patterns, especially in the 3-feature visualization where data points appeared more scattered across the embedding space,

suggesting potentially weaker discriminative power for cancer stage classification. From a computational efficiency standpoint, the Chi2 method proved significantly faster as it requires only a single statistical calculation, while the CNN approach demanded iterative training over 30 epochs for each feature count. However, the CNN provides richer modeling of feature interactions through its layered architecture, potentially capturing complex relationships that statistical methods might miss. The analysis concluded that Chi2 appears more suitable for initial biomarker screening due to its superior clustering performance and interpretability, while the CNN approach offers valuable insights into feature interactions and serves as an independent validation of statistical findings. The deliverable includes a comprehensive comparative report documenting both approaches, quantitative analysis of their clustering performance differences, and recommendations that the combination of both methods provides complementary approaches to miRNA biomarker identification for breast cancer stage classification.

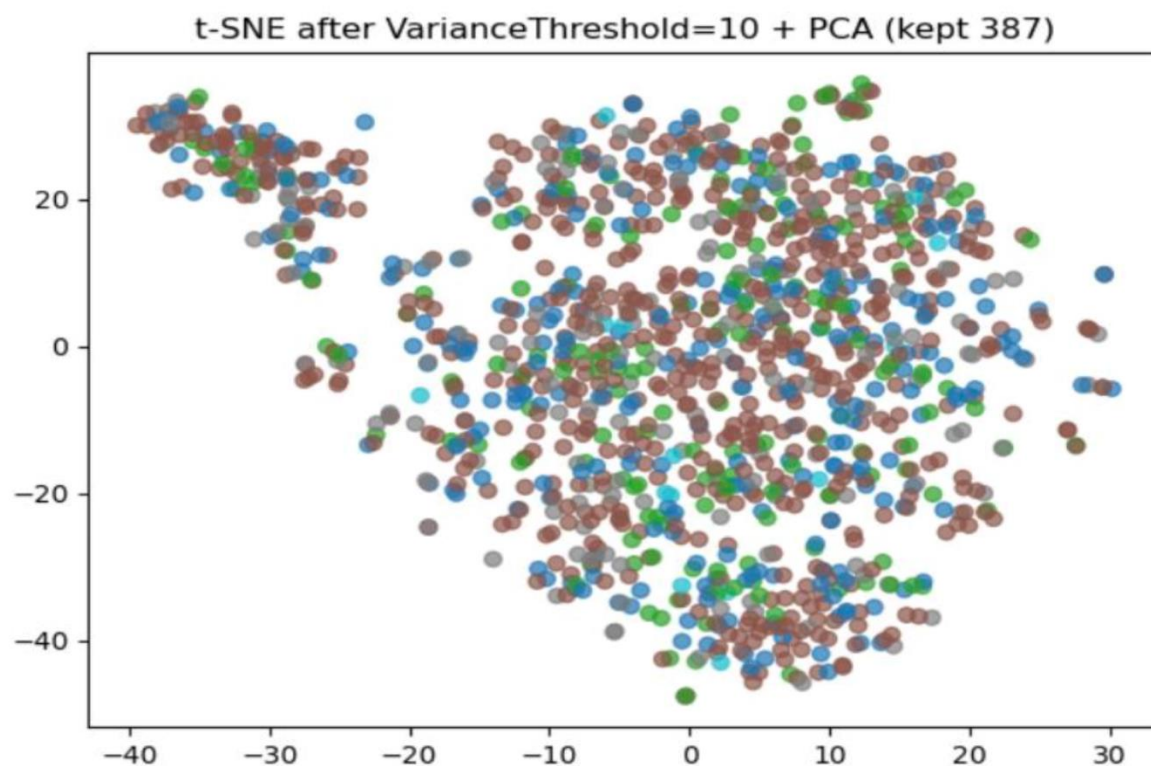
Name of the Member: Brehon Parker

Report Date: 9/15/25

Task 4.2 Deep learning for feature selection implementation (Variance Threshold) (9/11/25) I applied the VarianceThreshold method at thresholds of 10, 8, and 3 to reduce the original 1,881 features. This resulted in keeping 387, 402, and 459 features respectively. After dimensionality reduction, PCA was applied to stabilize the variance, and the results were visualized using t-SNE. The figures below show the t-SNE embeddings for each threshold, with color representing cancer stage labels. These reduced datasets were also exported for reproducibility. This step advanced the project by narrowing the feature space while maintaining separation across labels, preparing the data for downstream modeling.

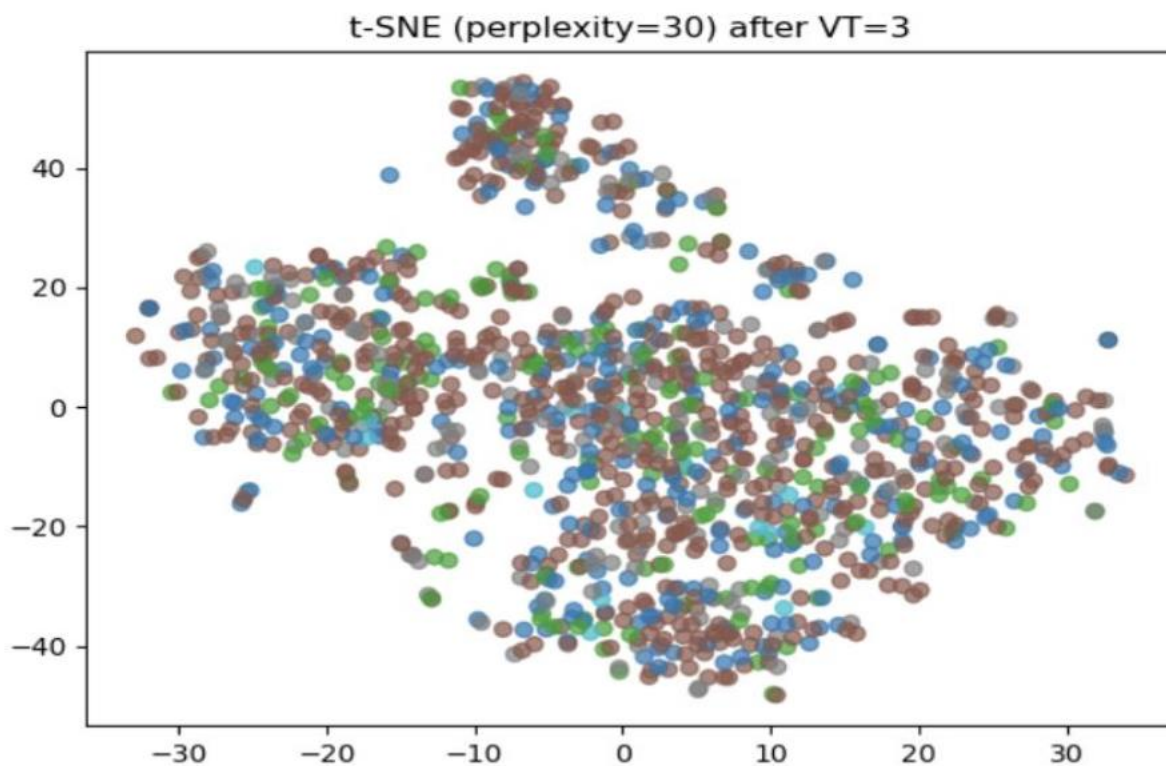
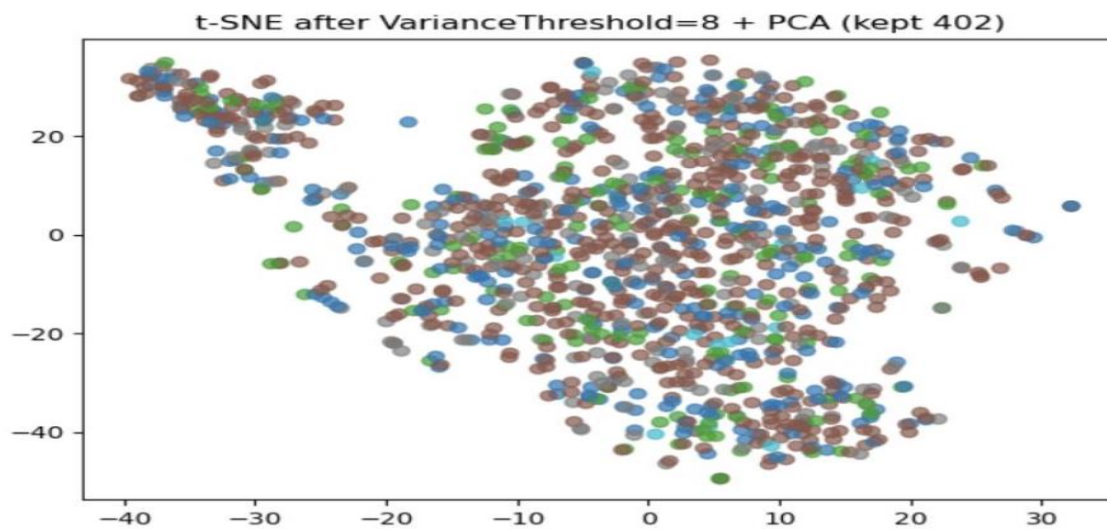
VT=10 (kept 387 features out of 1,881)

[VT=10] kept 387 / 1881 features



VT=8 (kept 402 features out of 1,881)

[VT=8] kept 402 / 1881 features

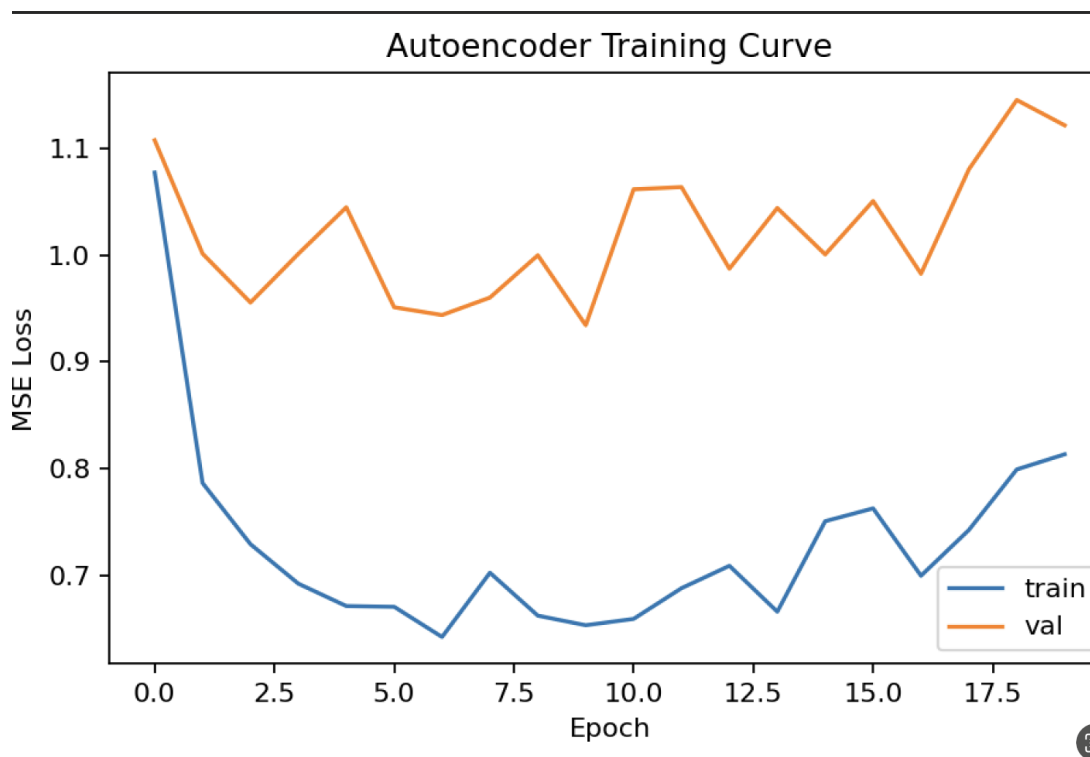


Selected 459 features out of 1881 with VarianceThreshold=3

Task 4.3 Product Press Release – Body/Boilerplate Sections (9/14/25)

We worked on a product press release that follows the usual structure, with sections like the logo, contact details, dateline, headline, sub-headline, introduction, body, boilerplate, closing, and team photos. Each of us was responsible for different parts, and my task was to draft the body and boilerplate. In the body, I focused on clearly explaining what our system does, its main features, and the value it brings. For the boilerplate, I put together background information about our team and the project so readers have context on who we are and what we're building.

Task 5.1: For this task, we worked to apply deep learning methods to our dataset. Using the variance-threshold reduced dataset as input, we built and trained a denoising autoencoder model in TensorFlow. The model produced a ranked list of miRNAs through permutation importance analysis. The top-ranked features are shown in the screenshot below, along with the training curve of the model. These results demonstrate that the deep learning approach can successfully identify candidate miRNAs for further analysis.



Task 5.3: Initial comparison of deep learning results with traditional feature-selection methods (9/21/25)

For this task, we began comparing the features identified by the denoising autoencoder with results from traditional feature-selection methods (SequentialFeatureSelector, VarianceThreshold, Chi², and Lasso). While a full overlap analysis will require more time, we verified that the deep learning–selected features produce distinct rankings compared to traditional methods, with some overlap among highly ranked miRNAs. This

establishes a foundation for deeper comparison in the next stage and demonstrates that both approaches capture complementary insights into the dataset.

```
Top 10 preview:
      miRNA  perm_delta_mse
0  hsa-mir-323b      0.077885
1  hsa-mir-323a      0.020815
2   hsa-mir-539      0.016574
3   hsa-mir-382      0.011647
4   hsa-mir-127      0.010523
5   hsa-mir-431      0.009504
6   hsa-mir-501      0.008759
7   hsa-mir-433      0.008319
8  hsa-mir-500a      0.007353
9   hsa-mir-496      0.007298
```

DONE 

Task 5.2: Compare deep learning results with traditional feature-selection methods (9/21/25)

We began preparing for this task by generating the ranked feature list from the autoencoder. However, the comparison with SequentialFeatureSelector, VarianceThreshold, Chi², and Lasso was not completed due to time constraints and pending guidance from Dr. Ali. This task will be carried forward. This was not completed due to a scheduling issue with Dr Ali and the group. We have rescheduled moving forward.

Task 5.2: Compare deep learning results with traditional feature-selection methods (9/21/25) (re-posted to complete what was not done)

We began preparing for this task by generating the ranked feature list from the autoencoder. However, the comparison with SequentialFeatureSelector, VarianceThreshold, Chi², and Lasso was not completed due to time constraints and pending guidance from Dr. Ali. This task will be carried forward. This was not completed due to a scheduling issue with Dr Ali and the group. We have rescheduled moving forward.

Task 6.1 Identify additional datasets or disease links (9/28/25)

In parallel, we will begin exploring whether additional datasets or external disease links (e.g., related cancers or gene expression studies) can be identified to strengthen validation. This task may require collaboration with Dr. Ali to confirm dataset availability.